

UTM TENSILE REPORT — UTM_Test_20260905_160536

DIC gauge strain (ENGINEERING: $\Delta L/L_0$) · engineering stress (F/A_0) · anchor self-calibration

43.0 MPa

UTS

4.1 MPa

Yield σ_y

2.01 GPa

Modulus E

1.4 %

Failure ϵ_f

601 kJ/m³

Toughness

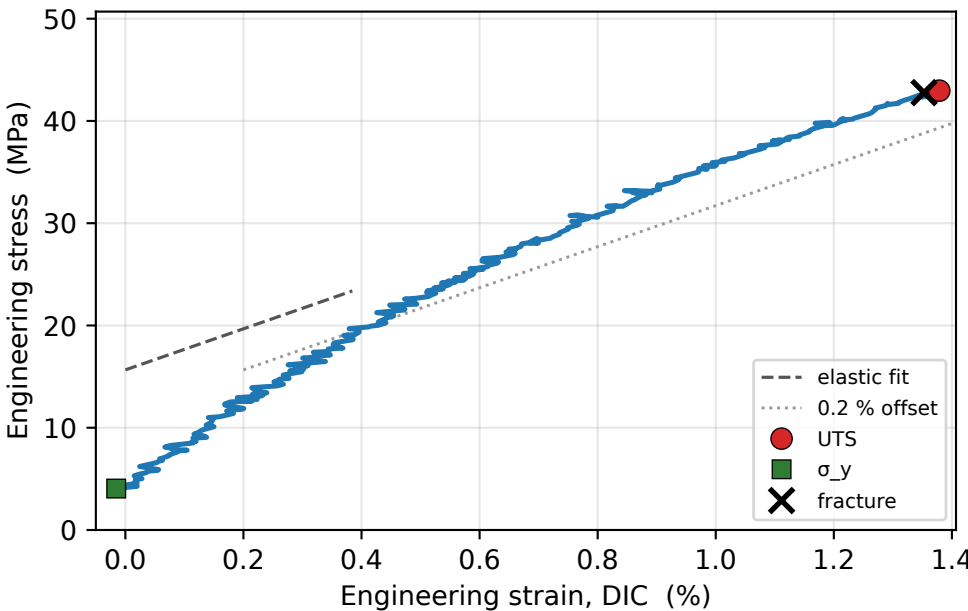
319 N

Preload anchor

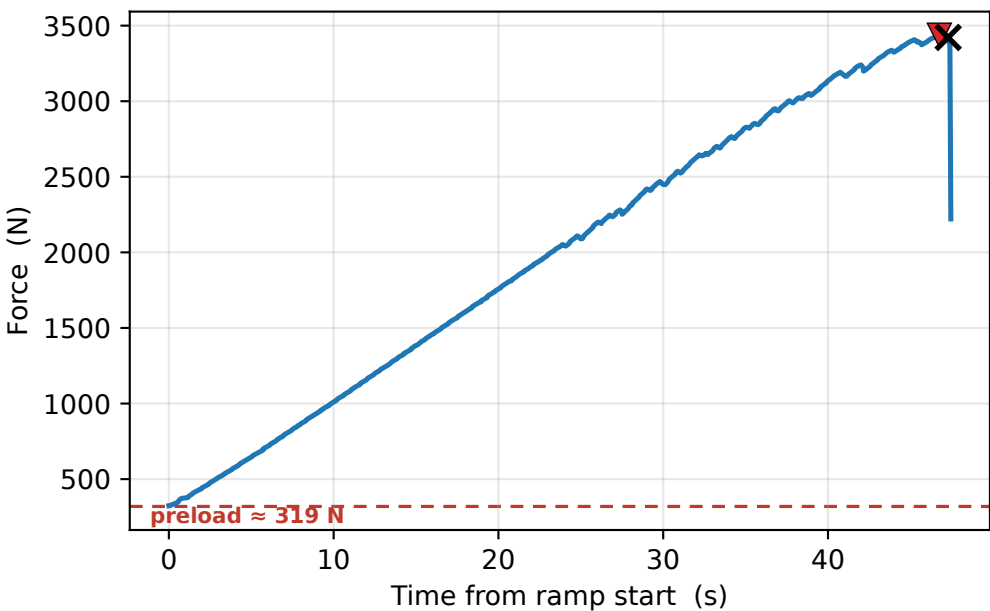
TEST SETTINGS

Specimen	UTM_Test_20260905_160536
DIC mode	White
Preload target	300 N
Test speed	0.100 mm/s
Cross-section	80 mm ²
Gauge length L ₀	80 mm
Infill (label)	100 %
Force calib	sc -0.0065 / off 320.112
Comment	—
Test date	2026-09-05 16:04:11
Duration	92.7 s

Stress-strain



Load vs time

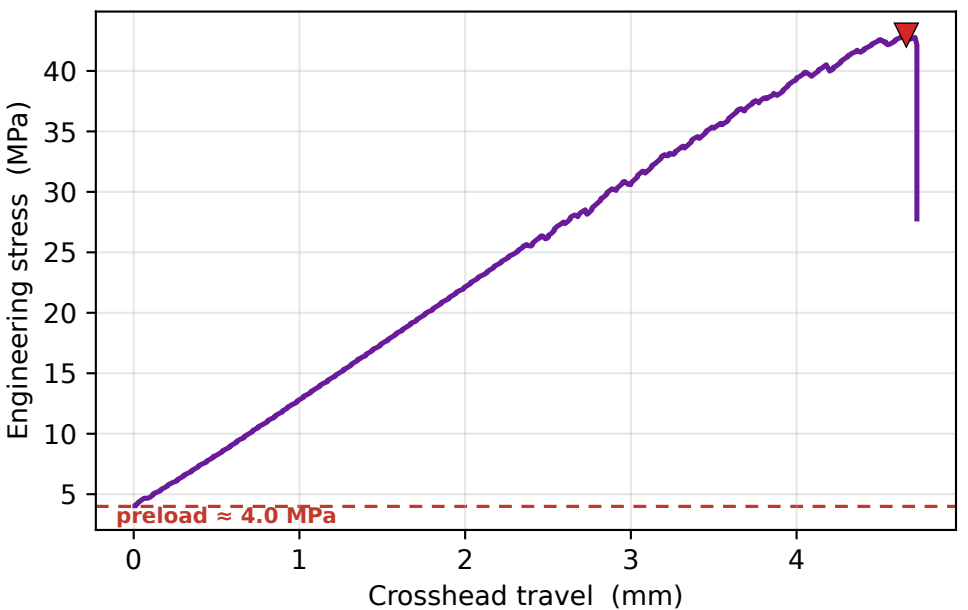


VALIDATION vs REFERENCES

prop	meas	Chacón	E-PLA k
UTS	43.0	32-60 PASS	×1.35
σ_y	4.1	30-50 low	×14.32
E	2.0	3-5.5 low	×1.43
ϵ_f	1.4 %	(range 3-7 %)	×5.91

⚠ strength outside journal range

Stress vs crosshead



DIC strain vs crosshead — the compliance gap

